

Fute Portal Backend

Architecture and current operating status of the self-hosted backend, following its migration off Firebase onto MongoDB on the temp server at 192.168.1.23.

Generated 2026-09-03 Node 24 · Express 4 · MongoDB 27017 Host: FS123 (192.168.1.23)

MIGRATION IN NUMBERS

What moved

27 Firestore collections migrated	48 User accounts	39 Employee records	30 Backend files touched
0 Cloud dependencies remaining			

AT A GLANCE

Current status

- Auth (bcrypt + JWT)

● CRUD & queries

● Pagination

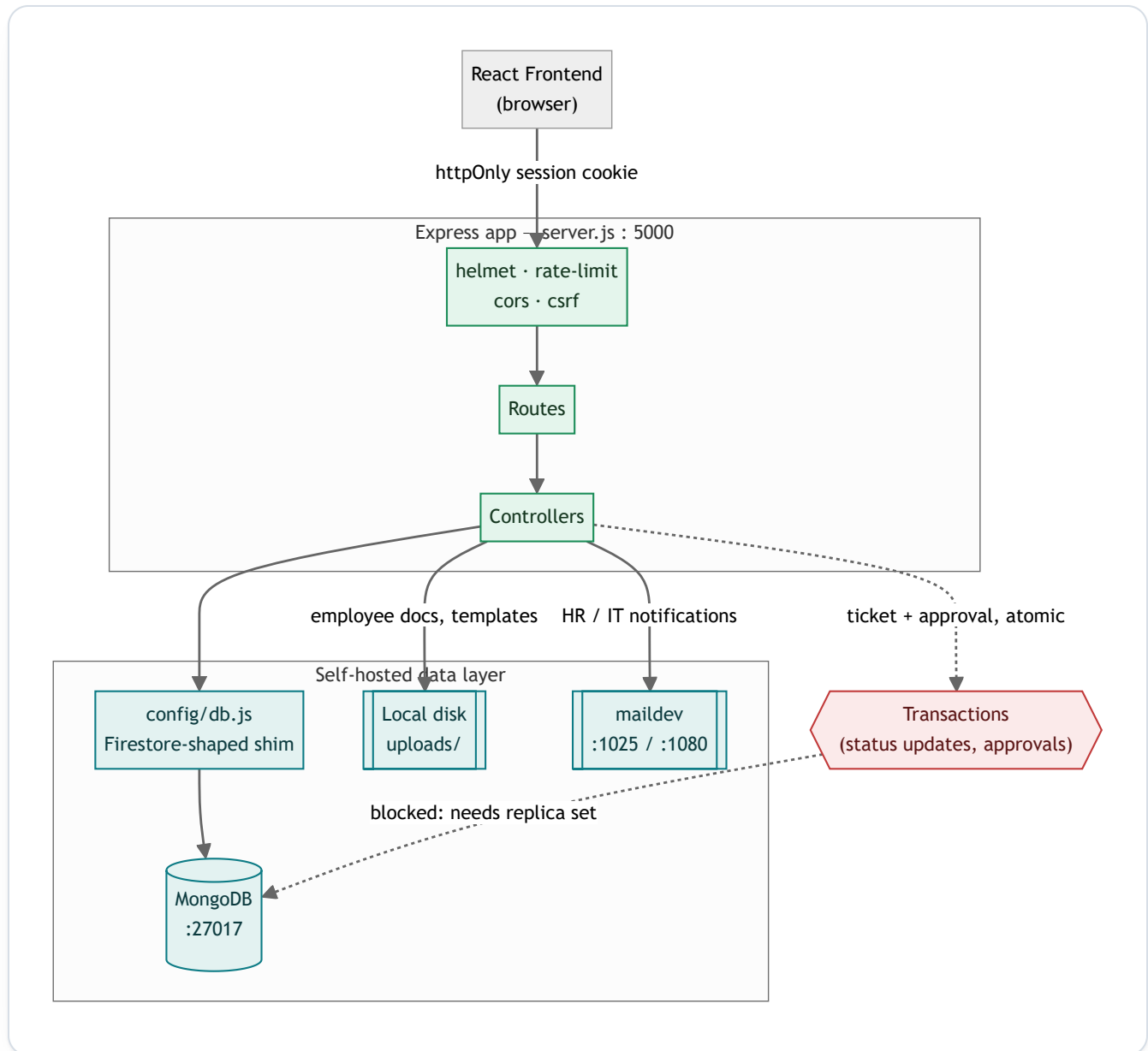
● Local file storage

● Email capture (maildev)

● Multi-doc transactions

● Process persistence

Request path



SEQUENCE

Life of a request

01 Cookie arrives

The browser attaches its httpOnly `fute_token` automatically — never read or set from page JavaScript.

02 Middleware chain

`helmet`, a 300-req/15-min rate limiter, an origin-checked `cors`, then `csrfMiddleware` on any mutating verb.

03 Auth + session

`authMiddleware` verifies the JWT, re-checks the live profile (60s cache), and confirms the session hasn't been revoked.

04 Route → controller

Role/permission middleware gates the route before the controller runs, e.g. `role('hr', 'founder')`.

05 config/db.js shim

Controllers call `db.collection(x).where().orderBy().limit()` exactly as they did against Firestore — the shim translates that onto the native MongoDB driver underneath.

06 Response

A uniform `{ success, message, data }` envelope, with any mutation also updating `audit_logs`.

Representative collections

COLLECTION	HOLDS	DOCS
users	Profiles — role, department, permission overrides	48
_auth_credentials	bcrypt password hashes, kept separate from profile docs	48
employees	HR directory records, including document storage paths	39
it_complaints / hr_complaints	Support tickets	6
sessions	Refresh-token rotation, revocation state	350
sales_leads	Sales desk pipeline	30
audit_logs	Admin action trail	100
settings	SLA policy, system config — fixed single docs	2

NOT YET WORKING

Two open gaps

Multi-document transactions

Ticket status updates and approval decisions write to two or three collections atomically. MongoDB refuses this on a standalone instance — it needs to run as a (single-node) replica set. Fails cleanly with a 500, no partial writes.

Process persistence

Windows OpenSSH kills PM2's daemon when the SSH session closes, since webteam has no interactive login. Fix is installing PM2 as a proper Windows Service — pending admin action.

CONFIGURATION

Environment

MONGODB_URL	mongodb://192.168.1.23:27017
MONGODB_DB_NAME	fute_portal
JWT_SECRET	96-byte random secret, generated at deploy time
SMTP_HOST / PORT	localhost : 1025 — maildev, no auth
PORT	5000
FRONTEND_URL	http://192.168.1.23 — CORS allow-list